

Structured Questions

Inorganic & Organic Chemistry Core Practicals

Hydrolysis of Halogenoalkanes / Chlorination of 2-Methylpropan-2-ol / Propan-1-ol Oxidation / AS Qualitative Analysis

Easy (2 questions)	/20
Medium (5 questions)	/64
Total Marks	/84

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Easy Questions

1 (a) This question is about some compounds of strontium.

State a test for the strontium cation, giving the expected result.

(2 marks)

(b) An unlabelled bottle was thought to contain solid strontium chloride.

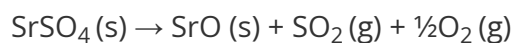
A sample of the solid was dissolved in distilled water for tests to identify the anion.

Complete the table to give the expected results of the anion tests.

Reagent added for test	Expected result for the strontium chloride solution
Barium chloride acidified with hydrochloric acid	
Silver nitrate acidified with nitric acid	

(2 marks)

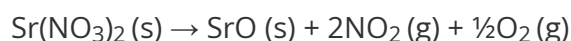
- (c) Anhydrous strontium sulfate undergoes thermal decomposition at approximately 1300°C.



Suggest why this decomposition is unlikely to be possible in a school laboratory.

(1 mark)

- (d) Anhydrous strontium nitrate decomposes at 570°C.



i) Describe how to ensure the strontium nitrate decomposes fully.

(1)

ii) State the colour of nitrogen dioxide gas.

(1)

iii) Give the test for oxygen and the expected positive result.

(1)

iv) The solid residue from the decomposition was added to distilled water.

Give **one** observation for the reaction that takes place, identifying the product of the reaction by name or formula.

(2)

Observation

Product

(5 marks)

2 (a) The white solids sodium sulfate and potassium carbonate may be distinguished using a flame test.

i) Identify a material from which the flame test wire could be made.
Justify your answer.

(2)

ii) Describe how to carry out a flame test on a solid, giving the expected flame colour for each of these compounds.

(4)

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(6 marks)

(b) Sodium sulfate and potassium carbonate may also be distinguished using **chemical** tests.

Give a **chemical** test for each compound which would confirm the identity of the **anion**.
Include the expected results.

Test 1 Test 2

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(4 marks)

Medium Questions

- 1 (a)** A student was provided with five test tubes labelled **A**, **B**, **C**, **D** and **E**, each containing a colourless aqueous solution.

The five solutions were known to be

barium chloride

nitric acid

potassium bromide

silver nitrate

sodium carbonate

The student carried out a series of tests to identify which test tube contained which solution.

- i) The student tested each solution using universal indicator paper. Only solution **A** turned the paper red.

Identify solution **A**.

(1)

- ii) The student mixed 1 cm³ of solution **A** separately with 1 cm³ of each of the other solutions.

There was no change for three of the mixtures but effervescence was observed when solution **A** was added to solution **C**.

Identify solution **C**.

(1)

- iii) Write an **ionic** equation for the reaction between solution **A** and solution **C**. Include state symbols.

(2)

iv) The student then mixed 1 cm³ samples of the remaining solutions as shown in **Table 1**.

Solutions mixed	Observation
B and D	no change
B and E	cream precipitate
D and E	white precipitate

Table 1

Identify the three remaining solutions.

(3)

Solution **B**

Solution **D**

Solution **E**

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(7 marks)

(b) Three of the cations in the compounds in (a) can be identified using flame tests.

Complete **Table 2**

Cation formula	Flame colour

Table 2

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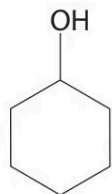
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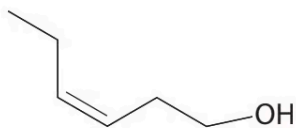
(3 marks)

2 (a) This question is about the reactions of three compounds with the formula $C_6H_{12}O$.

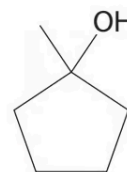
The compounds are cyclohexanol, Z-hex-3-en-1-ol and 1-methylcyclopentanol.



cyclohexanol



Z-hex-3-en-1-ol



1-methylcyclopentanol

Give a chemical test to show the presence of the $-OH$ group in all three compounds, including the expected result.

(2 marks)

(b) i) Give a chemical test to show the presence of the carbon-carbon double bond in Z-hex-3-en-1-ol, including the expected result.

(2)

ii) The test you have given in (b)(i) is repeated with 1-methylcyclopentanol.

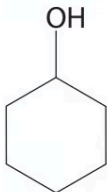
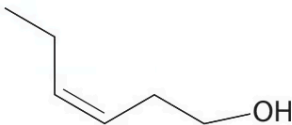
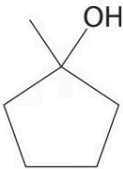
Give the observation for this test with 1-methylcyclopentanol.

(1)

(3 marks)

(c) Separate samples of each of these compounds are warmed with acidified potassium dichromate(VI).

Complete the table to give the colour changes observed, if any.

Compound	Colour change
	
	
	

(2 marks)

- (d) Spectroscopy provides information about the structure of these three compounds. Some infrared data is given in the table.

Group	Wavenumber range/ cm^{-1}
O–H stretching in alcohols	3750–3200
O–H stretching in carboxylic acids	3300–2500
C = O stretching in aldehydes	1740–1720
C = O stretching in ketones	1720–1700
C = O stretching in carboxylic acids	1725–1700
C–H stretching in aldehydes	2900–2820 2775–2700
C = C stretching in alkenes	1669–1645

i) Identify the wavenumber range and the bond responsible for **one** peak which you would expect to see in the infrared spectra of all three compounds.

(1)

ii) Identify the wavenumber range and the bond responsible for **one** peak which you would expect to see in the infrared spectra of only one of the compounds.

(1)

iii) Give a reason why there is a peak at $m/z = 100$ in the mass spectra of all three compounds.

(1)

iv) Fragmentation of 1-methylcyclopentanol results in a significant peak at $m/z = 85$.

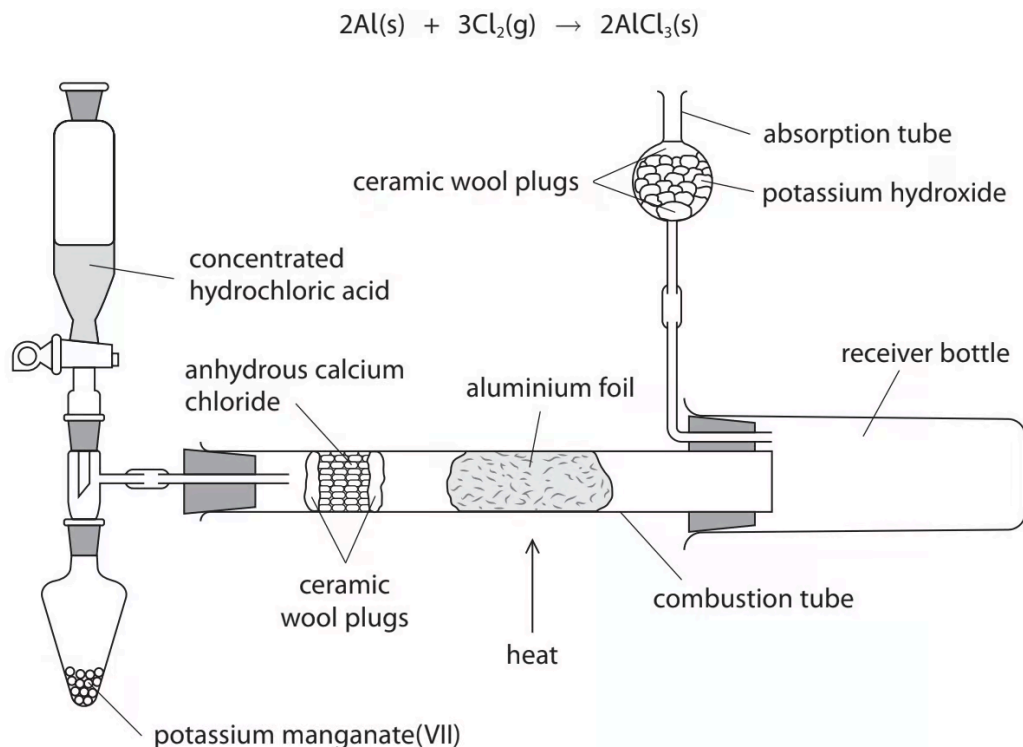
Suggest the structures of the **two** species formed when one bond in 1-methylcyclopentanol breaks resulting in the peak at $m/z = 85$.

(2)

(5 marks)

- 3 (a)** This question is about the preparation of anhydrous aluminium chloride, AlCl_3 , which reacts vigorously with water and must be stored in tightly sealed containers.

A sample of anhydrous AlCl_3 was prepared by passing chlorine gas over hot aluminium foil using the apparatus shown.



Step 1	Assemble the apparatus with about 5 g of potassium manganate (VII) in the pear-shaped flask, 10 cm ³ of concentrated hydrochloric acid in the tap funnel and a known mass of aluminium foil in the combustion tube.
Step 2	Carefully open the tap of the funnel, allowing the acid to enter the pear-shaped flask drop by drop. Wait for twenty seconds.
Step 3	Heat the aluminium foil until it glows brightly. Continue heating until the reaction is complete. Allow the apparatus to cool before closing the tap of the funnel.
Step 4	Remove the receiver bottle, quickly scrape the product into a sample tube and seal with a lid.

Granules of anhydrous calcium chloride are held between two ceramic wool plugs in the combustion tube.

i) Explain the purpose of the anhydrous calcium chloride.

(2)

ii) Give the reason why granules of anhydrous calcium chloride are used rather than powder.

(1)

(3 marks)

(b) The reaction occurring in Step 2 produces chlorine gas.

i) Identify the main hazard related to chlorine gas, giving the **best** way of minimising the risk when using this gas.

(2)

ii) Give a reason why the concentrated hydrochloric acid is added 'drop by drop' to the pear-shaped flask. (1)

(3 marks)

(c) Suggest why the heating of the aluminium in Step 3 is delayed by 20 s after the initial production of chlorine gas.

(1 mark)

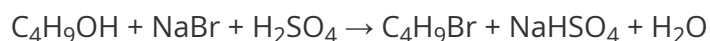
(d) State how you would know the reaction is complete in Step 3.

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(1 mark)

(e) Suggest the purpose of the potassium hydroxide in the absorption tube.

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(1 mark)

- 4 (a)** A sample of 1-bromobutane may be prepared by reacting butan-1-ol with sodium bromide and 50% concentrated sulfuric acid.



Procedure

Step 1	Add suitable quantities of butan-1-ol and sodium bromide solution to a round-bottom flask. Place the flask in a cold water bath. Add concentrated sulfuric acid drop by drop to the flask.
Step 2	Heat the mixture in the flask under reflux for about 45 minutes.
Step 3	Rearrange the apparatus for distillation and distil the reaction mixture. The distillate collected contains 1-bromobutane and water in separate layers. Remove as much of the water layer as possible.
Step 4	Transfer the impure 1-bromobutane to a separating funnel, add sodium hydrogencarbonate solution and shake the mixture. Run off the organic layer into a clean conical flask.
Step 5	Add anhydrous calcium chloride, stopper the flask and allow it to stand. Decant the liquid.
Step 6	Distil the product over a suitable temperature range to give pure 1-bromobutane.

Data

Property	Butan-1-ol	1-Bromobutane
Density / g cm ⁻³	0.810	1.27
Molar mass / g mol ⁻¹	74	137
Boiling temperature / °C	118	102

Suggest why the percentage yield of 1-bromobutane might be lower if the cold water bath was **not** used in Step 1.

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 (2 marks)

(b) i) State what must be added to the mixture in the flask before heating in Step 2.

(1)

ii) Draw a labelled diagram of the apparatus that you would use to heat the mixture under reflux in Step 2.

(3)

.....

 (4 marks)

(c) Purification of the product occurs in Steps **3–6**.

i) State why sodium hydrogencarbonate solution is added in Step **4**.

(1)

ii) Addition of sodium hydrogencarbonate solution in Step **4** causes vigorous effervescence.

Explain how the problem associated with Step **4** should be dealt with.

(2)

iii) Give the purpose of the anhydrous calcium chloride used in Step **5**.

(1)

iv) State how the appearance of the organic liquid would change in Step **5**.

(1)

(5 marks)

(d) For the final distillation in Step **6**, a thermometer with a scale giving readings to the nearest 1°C was provided. .

Give a suitable temperature **range** for the collection of the pure 1-bromobutane.

(1 mark)

(e) A student was asked to prepare 20 cm³ of 1-bromobutane using the procedure described. The student knew that the percentage yield would be less than 100%.

i) Give **one** possible reason for the yield being less than 100%.

(1)

ii) After some research the student decided to use 21.0 g of butan-1-ol to prepare 20 cm³ of 1-bromobutane.

Calculate the percentage yield that the student expected to obtain.

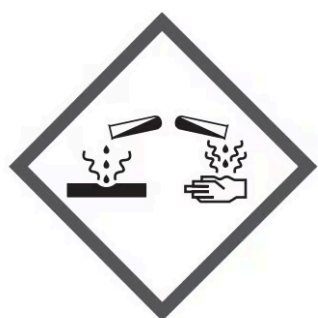
(4)

(5 marks)

5 (a) The halogenoalkane 2-chloro-2-methylpropane may be prepared from 2-methylpropan-2-ol. **Procedure**

Step 1	Add 35 cm ³ of concentrated hydrochloric acid to 8.00 g of 2-methylpropan-2-ol in a conical flask. Swirl the mixture gently for 20 minutes.
Step 2	Two distinct layers form. The upper (organic) layer contains the required product. The lower aqueous layer is removed using a separating funnel.
Step 3	Add a solution of sodium hydrogencarbonate to the organic layer. Swirl gently. Stopper the separating funnel and shake it. Invert the separating funnel and open the tap.
Step 4	Return the separating funnel to its upright position, remove the stopper and run off the aqueous layer. Transfer the organic layer into a clean conical flask.
Step 5	Add some anhydrous sodium sulfate. Leave the flask to stand and decant off the liquid.
Step 6	Distil the liquid, collecting the product between 50°C and 52°C.

i) The concentrated hydrochloric acid used in Step 1 was labelled



Suggest **two** safety precautions, other than wearing safety spectacles and a laboratory coat, to minimise the risk when using this reagent in Step 1.

(2)

ii) Explain why the product in the organic layer in Step 2 does not mix with the aqueous layer.

(2)

iii) State why the tap of the separating funnel must be opened in Step 3.

(1)

iv) State why anhydrous sodium sulfate is added to the organic layer in Step 5.

(1)

v) Draw the apparatus required to distil the product and collect the distillate between 50°C and 52°C in Step 6.

(4)

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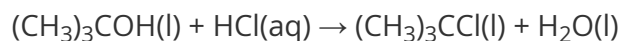
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(10 marks)

(b) The equation for the reaction is



The final product after distillation weighed 2.62 g.

Calculate the percentage yield.

(3 marks)

(c) The haloalkane produced is used in an experiment to compare its rate of hydrolysis with two other halogenoalkanes.

A student dissolves separate 1.0 cm³ samples of each halogenoalkane in ethanol and adds 2 cm³ of silver nitrate solution.

The time taken for a precipitate to form is recorded. The results are shown.

Halogenoalkane	Time / s
2-chloro-2-methylpropane	5
1-chloro-2-methylpropane	320
1-bromo-2-methylpropane	140

The student concludes that both the structure of the halogenoalkane and the identity of the halogen affect the rate of hydrolysis.

Explain how the results support this conclusion.

(3 marks)